

Safety Enforcement in PSPO: --freq update versus --freq 1

1 The source of the distinction

PPO alternates between two operations: it first collects a rollout into a buffer, and it then trains on that buffer. The training operation is not one large gradient update. PPO shuffles the rollout buffer, divides it into minibatches, and performs an optimizer step after every minibatch. It normally passes over the same rollout buffer for several epochs. The two PSPO frequency settings differ in whether safety is enforced after each of these optimizer steps or once after the complete training operation.

Let a rollout buffer contain

$$\mathcal{D} = \{(s_t, a_t, r_t, \hat{A}_t, \log \pi_{\theta_{\text{old}}}(a_t \mid s_t))\}_{t=1}^N, \quad (1)$$

where $\hat{A}_t$ is the estimated advantage and θ_{old} denotes the actor that generated the rollout. The old log-probabilities and advantages are held fixed while PPO trains on this buffer.

2 Minibatch policy-gradient computation

Suppose the current minibatch is $M_j \subset \mathcal{D}$ and the current actor parameters are θ_j . PPO forms the probability ratio

$$r_t(\theta_j) = \frac{\pi_{\theta_j}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \quad (2)$$

and the clipped minibatch objective

$$J_j(\theta_j) = \frac{1}{|M_j|} \sum_{t \in M_j} \min \left\{ r_t(\theta_j) \hat{A}_t, \text{clip}(r_t(\theta_j), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right\}. \quad (3)$$

The implementation also includes the value-function and entropy terms, but the clipped policy objective above is the important part for understanding the actor update. The minibatch gradient is

$$g_j = \nabla_{\theta} J_j(\theta) |_{\theta=\theta_j}. \quad (4)$$

Adam immediately uses this gradient to propose new actor parameters,

$$\bar{\theta}_{j+1} = \text{AdamStep}(\theta_j, g_j). \quad (5)$$

The gradient for the next minibatch is consequently evaluated at the actor produced by the preceding optimizer step; PPO does not normally accumulate all minibatch gradients and apply them only once at the end.

If the buffer contains N samples, the minibatch size is B , and PPO makes E epochs over the buffer, one rollout produces approximately

$$K = E \left\lceil \frac{N}{B} \right\rceil \quad (6)$$

optimizer steps. Thus one rollout and one policy update are not synonymous: one rollout is normally followed by many minibatch parameter updates.

3 Abstract safety-enforcement operation

Write Θ_{safe} for the certified-safe actor-parameter set and let $\mathcal{E}$ denote the PSPO enforcement operation. Conceptually,

$$\mathcal{E}(\bar{\theta}; \theta^{\text{safe}}) = \begin{cases} \bar{\theta}, & \text{if the proposal is certified or contained in a certified region,} \\ \text{Proj}_{\Theta^{\text{LID}}}(\bar{\theta}), & \text{if a certified local region is available,} \\ \theta^{\text{safe}}, & \text{if certification or projection cannot be completed.} \end{cases} \quad (7)$$

Here θ^{safe} is the last certified-safe snapshot and $\Theta^{\text{LID}} \subseteq \Theta_{\text{safe}}$ is a certified local parameter region. The verify-first and region-first variants construct this decision in different orders, but both are intended to return a certified-safe actor at an enforcement boundary. Safety enforcement applies to the actor parameters; the critic parameters need not be projected into this region.

4 --freq update: enforcement after every minibatch step

With `--freq update`, each Adam proposal is immediately followed by safety enforcement:

$$g_j = \nabla J_j(\theta_j), \quad (8)$$

$$\bar{\theta}_{j+1} = \text{AdamStep}(\theta_j, g_j), \quad (9)$$

$$\theta_{j+1} = \mathcal{E}(\bar{\theta}_{j+1}; \theta_j^{\text{safe}}) \in \Theta_{\text{safe}}. \quad (10)$$

The next minibatch gradient is computed at the enforced parameters θ_{j+1} . Therefore enforcement can change not only the final actor but also every subsequent minibatch gradient and the entire optimization path. At the end of the PPO training operation, all K intermediate actor iterates have passed through an enforcement boundary.

5 --freq 1: enforcement after one complete PPO training phase

With `--freq 1`, PPO first performs all K optimizer steps associated with the rollout buffer. Denoting the unenforced intermediate parameters by $\tilde{\theta}_j$, the inner sequence is

$$g_j = \nabla J_j(\tilde{\theta}_j), \quad (11)$$

$$\tilde{\theta}_{j+1} = \text{AdamStep}(\tilde{\theta}_j, g_j), \quad j = 0, \dots, K-1. \quad (12)$$

Only the aggregate candidate is then enforced:

$$\theta_{\text{next}} = \mathcal{E}(\tilde{\theta}_K; \theta^{\text{safe}}) \in \Theta_{\text{safe}}. \quad (13)$$

The intermediate actors $\tilde{\theta}_1, \dots, \tilde{\theta}_{K-1}$ need not belong to Θ_{safe} . They arise while PPO is training on an already-collected buffer, however, and are not used to start a new environment rollout. Enforcement occurs after the complete training phase and before the next rollout begins.

This explains the rollout terminology. Internally, `--freq 1` means “enforce after every one train phase.” In on-policy PPO there is one such training phase after each rollout collection, so it is described as “every one rollout.” The word describes the outer collect–train cycle, not the source of an individual gradient.

6 Direct comparison

	--freq update	--freq 1
Enforcement boundary	Every minibatch optimizer step	End of the complete PPO train phase
Enforcements per rollout	Approximately K	One
Intermediate actor iterates	Enforced after every step	May be uncertified during training
Next minibatch gradient	Evaluated at the enforced actor	Evaluated at the raw preceding Adam iterate
Actor used for next rollout	Certified-safe	Certified-safe

For example, if $N = 2048$, $B = 64$, and $E = 10$, PPO performs $K = 10(2048/64) = 320$ optimizer steps after one rollout. The `update` setting enforces safety 320 times, whereas the `1` setting enforces it once after the 320-step sequence.

7 Why the resulting policy is safe

For either setting, PSPO does not expose the completed training-phase result as the next certified policy until enforcement succeeds. A valid proposal is accepted or projected into a certified region; a failed enforcement restores the last certified-safe snapshot. PSPO also flushes any pending aggregate update before final evaluation and saving. Hence the actor at the next rollout boundary and the final saved actor belong to Θ_{safe} , subject to the soundness and coverage assumptions of the safety certificate.

This parameter-level guarantee should be distinguished from stochastic action sampling. A categorical policy can assign nonzero probability to an unsafe action even when its greedy action satisfies the certified ordering. During exploration, PSPO therefore additionally uses the action shield to replace unsafe sampled actions. The certificate protects the actor at enforcement boundaries, while the shield protects executed actions during rollout collection.